

اتحاد طلبة هندسة الطاقة والاتصالات

pce

حلول تجارب لاب إلكترونيات

Dont miss the change.. be part of it

رئيس الجمعية: علي الدويكات



Groups/pce.fet



PCE_FET

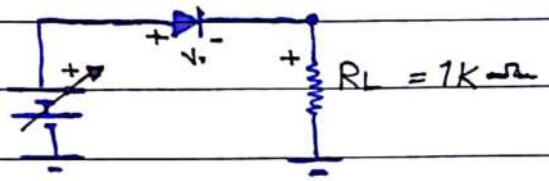
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Expirement One :-

General purpose Diode

Rocedure :

1. The Diode symbol (▶|) on the DMM is selected and the diode is tested, $V_\gamma = 0.558V$
2. The circuit is wired.



Forward =

$-E_S + V_D + V_R = 0$

$E_S = V_D + V_R$

$E_S = V_D + I_D R$

$\rightarrow E_S = 0.1$

$0.1 = 0.7 + I_D + 1000$

$I_D = -0.6\,mA$

3. In the table, Es is sat to the voltage (o-3) by increasing the dc supply voltage in small steps. for each step while the foltage across and shrink through the diode are measured. The results are recorded.

4. The diode is reversed. And again the voltage across and current through the diode are measured while (Es) is been sat to the voltage (0-3):

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Es	Forward		Reversed	
(V)	V_F (V)	I_F (mA)	V_R (V)	I_R (mA)
0	-0.013	0	-26.7	0
0.1	92.9mV	0	85.5mV	0
0.2	185.6mV	0	192.2	0
0.3	278.8mV	0.003	288.7	0
0.4	365.6 mV	0.018	0.391 V	0
0.5	0.42 V	0.071	0.490 V	0.05mA
0.6	0.448	0.135	0.587 V	0.05mA
0.7	0.471	0.218	0.688	0.06 mA
0.8	0.486	0.3	0.795	0.07
0.9	0.498	0.386	0.894	0.08
1.0	0.509	0.483	0.997	0.09
1.2	0.525	0.662	1.192	0.11
1.4	0.538	0.847	1.395	0.13
1.6	0.552	1.129	1.697	0.15
2.0	0.569	1.421	2 V	0.18
2.4	0.576	1.809	2.4 V	0.22
2.7	0.589	2.097	2.702	0.25
3.0	0.59	2.384	3.00	0.27

percentage error for V_D = $\frac{|V_D \text{ measured} - V_D \text{ calculated}|}{100}$ = 0.042 %

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percentage error for $I_D = \frac{10 + 0.61}{100} = 0.6\%$

5. V_D versus I_D is ploted on graph and down. Then the load line is drawn and the static Resistance R_S at ($E_S = 1.5V$) is determined.

Conclusion :-

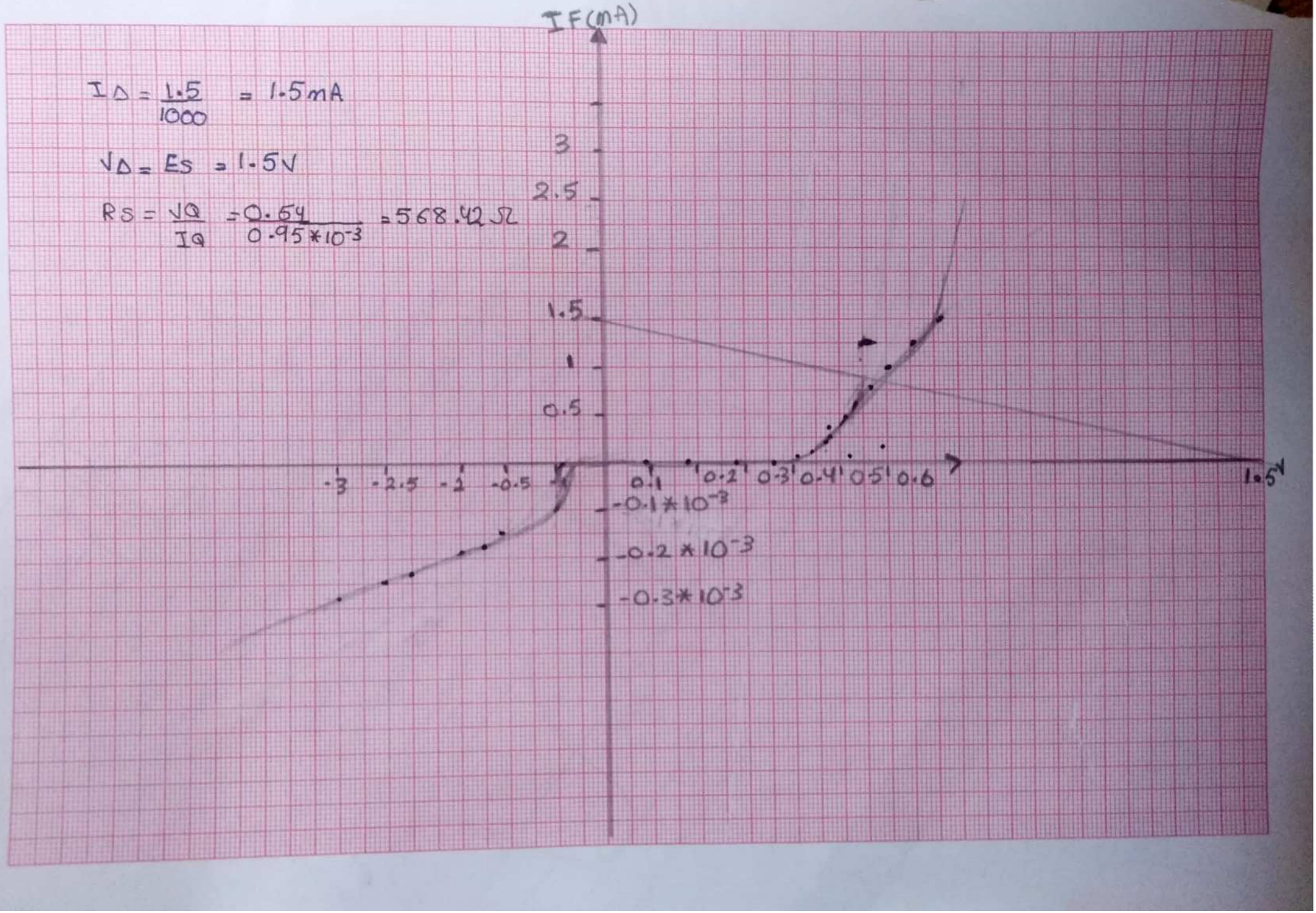
In this expirement we find the diode characteristics, to get diode the forward region we wired the positive polar at the supply to Anode terminal, and to get to reverse region we wined the positive polar to the cathode terminal.
Also, we find the static resistance using the law $R_S = \frac{V_{DQ}}{I_{DQ}}$, according to the characterstics graph.

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$I_D = \frac{1.5}{1000} = 1.5\,mA$

$V_D = E_S = 1.5\,V$

$R_S = \frac{V_Q}{I_Q} = \frac{0.54}{0.95 * 10^{-3}} = 568.42\,\Omega$



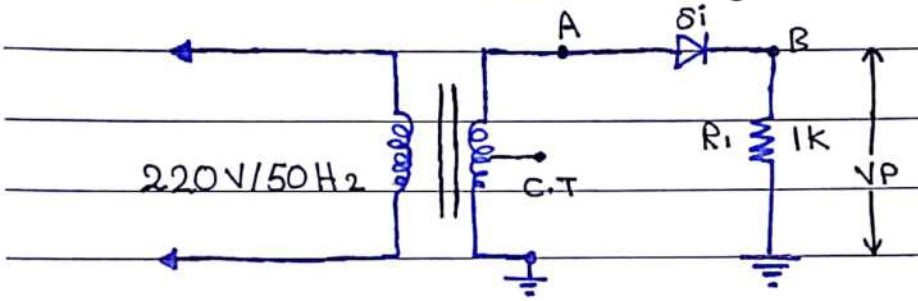
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Experiment :-

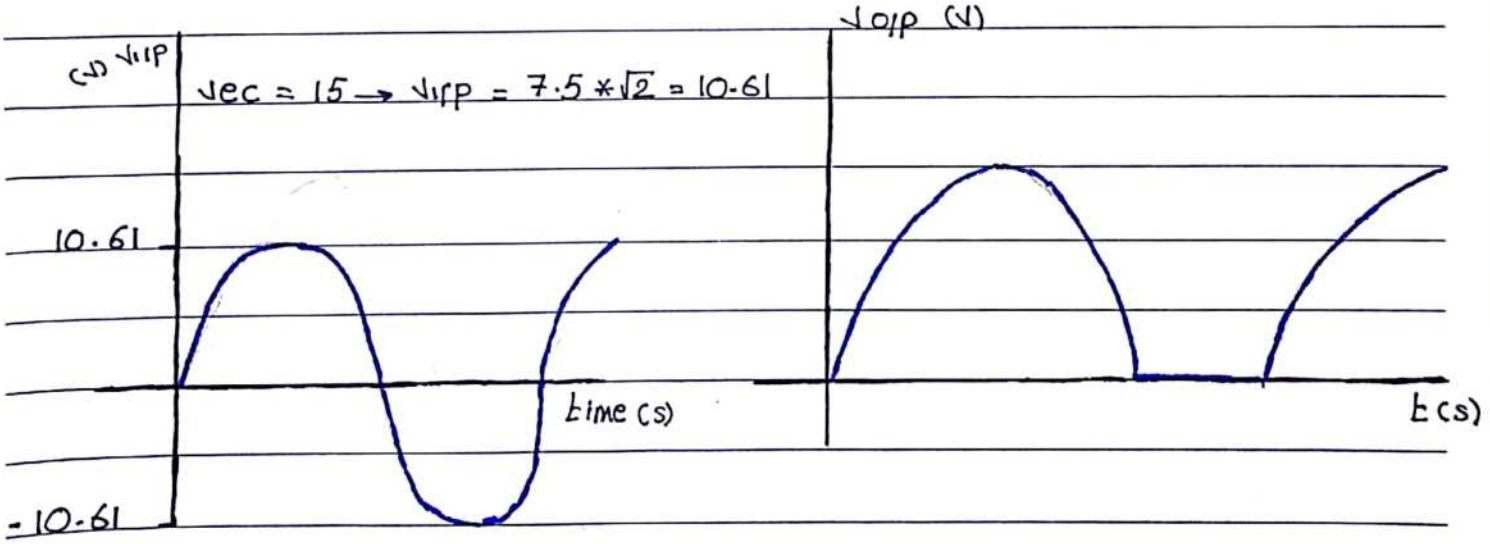
Diode Application (Rectifire circuits)

Half wave Rectifire :-

wire the circuit shown in figure :-



2. plug the transformer into mains. connect CH1 of the oscillascope at point A. measure and draw scale on the graph provided.



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3. set CH_2 to DC coupling and at point B, measure and draw V_{out}. calculate the avarage DC voltage (V_{pc}):

$$V_{DC} = \frac{V_{P(OIP)}}{\pi} = 3.2\ (V)$$

$$F_{out} = 50\ Hz$$

4. IF 220 μF capcitors is connected in parallel with R_L, calculate V_{DC}, V_{rip} and the ripple factor

$$V_{DC} = \left(1 - \frac{1}{2FR_C}\right) * V_{pout}$$

$$= \left(1 - \frac{1}{2*50*1000*100*10^{-6}}\right) * 10$$

$$= 9\ (V)$$

$$V_r = \frac{10}{50\ (10^3)\ (100)\ (10^{-6})} = 1V$$

$$V = \frac{V_r}{V_{DC}} = \frac{1}{9} = 0.11$$

4. IF the 220 μF is replaced by 4.7 μF and 100 μF. will the V_{DC}, V_{rip} and ripple factor be effected.

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yes, because the depend on C

$V_{DC} = \left(1 - \frac{1}{2FRC}\right) * V_{pout}$

C = 220 MF

V_{DC} = 9.56(V)

$r = \frac{V_r (P-P)}{V_{DC}}$

$V_r = \frac{V_p}{FR_LC}$

C = 4.7 MF

V_{DC} = 11.3 (V)

5. The circuit was cuired and checked supervisor.

6. Using the osillscope measure ,Vin at A , Vpout at point B, VDC.

7. The transformer was unpluged ,220 MF capacitor was connected in parralel with RL.

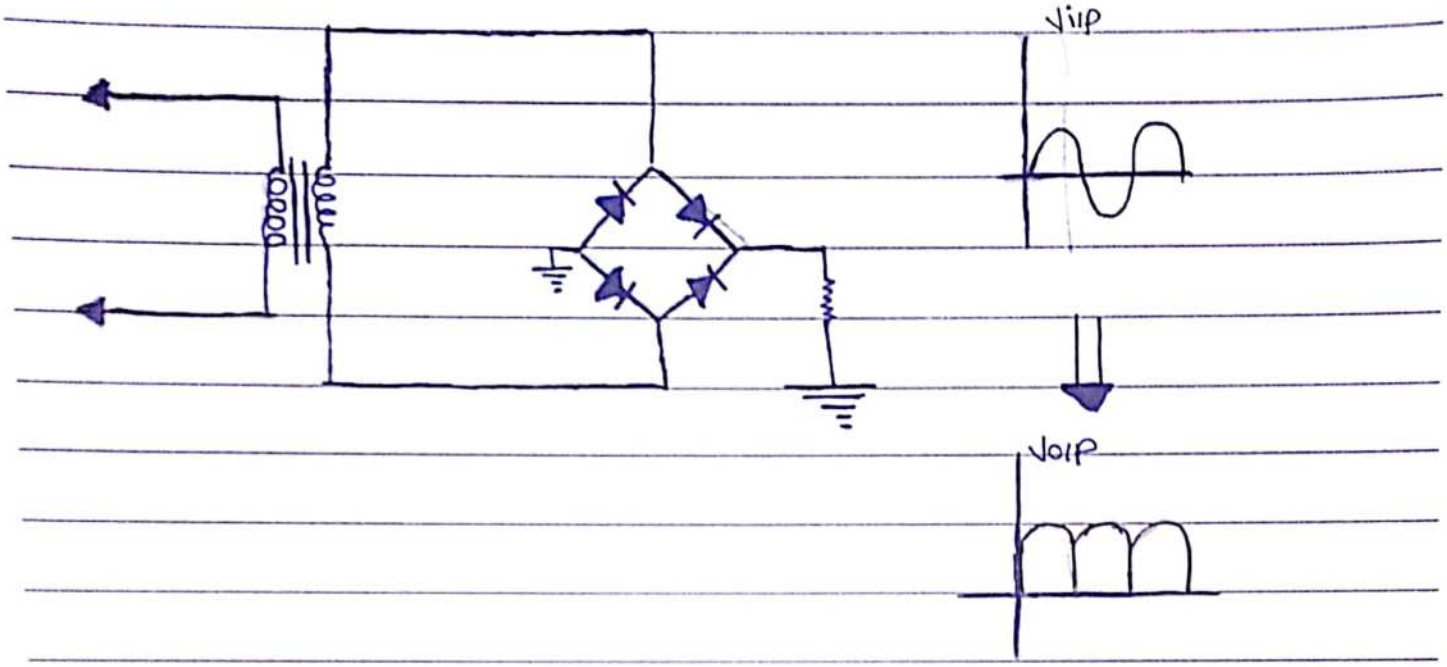
8. VDC , Vripp are measured.

9. step 7,8 are reapeted for 100 MF ,4.7 MF capicitors.

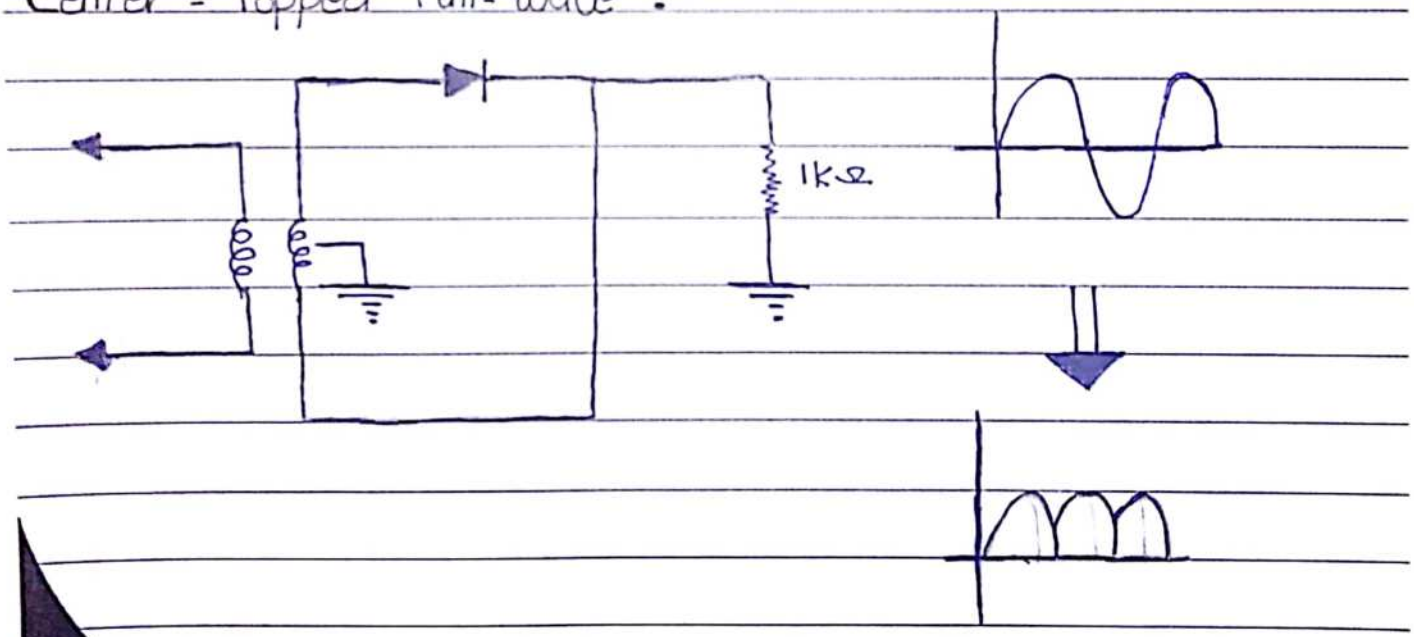
10. Results are recorded.

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Bridge Full-wave Rectifier :-



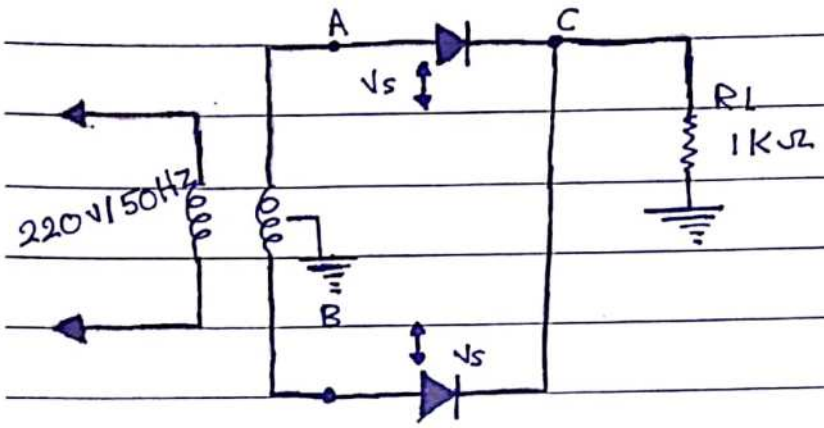
Center-Topped Full-wave :-



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Tapped Full wave Rectifier :-

Circuit is wired and steps are repreated.



$V_{p\,out} = 10V$

$V_{DC} = \frac{2*10}{\pi} = 6.37\ V$

$F_{out} = 2*50 = 100\ Hz$

with filter $C = 100\ MF$

$V_{DC} = \left(1 - \left(\frac{1}{2*100*1000*100*10^{-6}}\right)\right) * 10 = 9.5\ (V)$

$V_r = \frac{10}{100*10^3*100*10^{-6}} = 0.5\ V$

$r = \frac{0.5}{9.5} = 0.053$

$C = 220\ MF$

$\rightarrow V_{DC} = 9.77\ (V)$

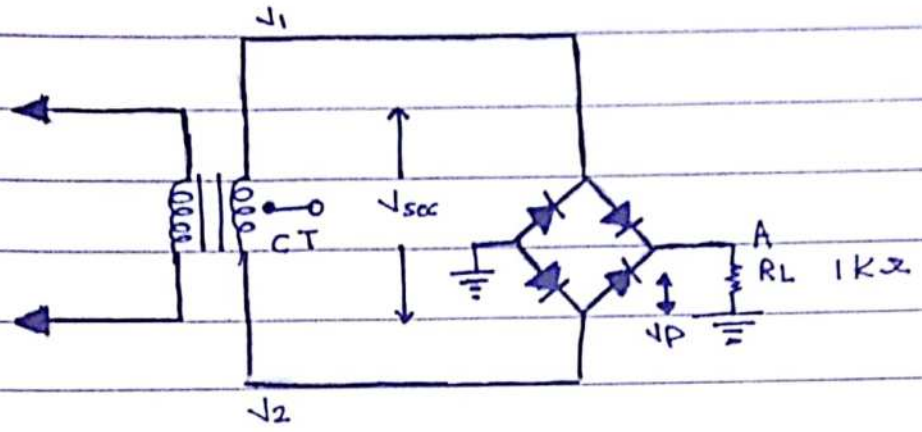
$C = 4.7\ MF$

$\rightarrow V_{DC} = 0.64$

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Bridge Full-wave Rectifier :-

1. The circuit is wired and steps are repeated.



$V_{po} = 10.61 - 1.4 = 9.2$ (V)

$V_{DC} = \frac{2 * 9.2}{\pi} = 5.9$ V

F = 100 Hz

with filter C = 100 MF

$V_{DC} = \left(1 - \left(\frac{1}{2 * 100 * 10^3 * 10 * 10^{-6}}\right)\right) * 9.2 = 8.74$ (V)

$V_r = \frac{9.2}{100 * 10^3 * 220 * 10^{-6}} = 0.46$ (V)

$r = \frac{0.46}{8.74} = 0.053$

$V_{DC} = \left(1 - \left(\frac{1}{2 * 100 * 10^3 * 220 * 10^{-6}}\right)\right) * 9.2 = 8.99$

$V_r = 0.21$

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$$r = \frac{V_r}{V_{DC}} = \frac{0.21}{8.99} = 0.025$$

$C = 4.7\ \mu F$

$$V_{DC} = \left(1 - \left(\frac{1}{2 * 100 * 10^3 * 4.7 * 10^{-6}}\right)\right) * V_{D\,out} = 9.79$$

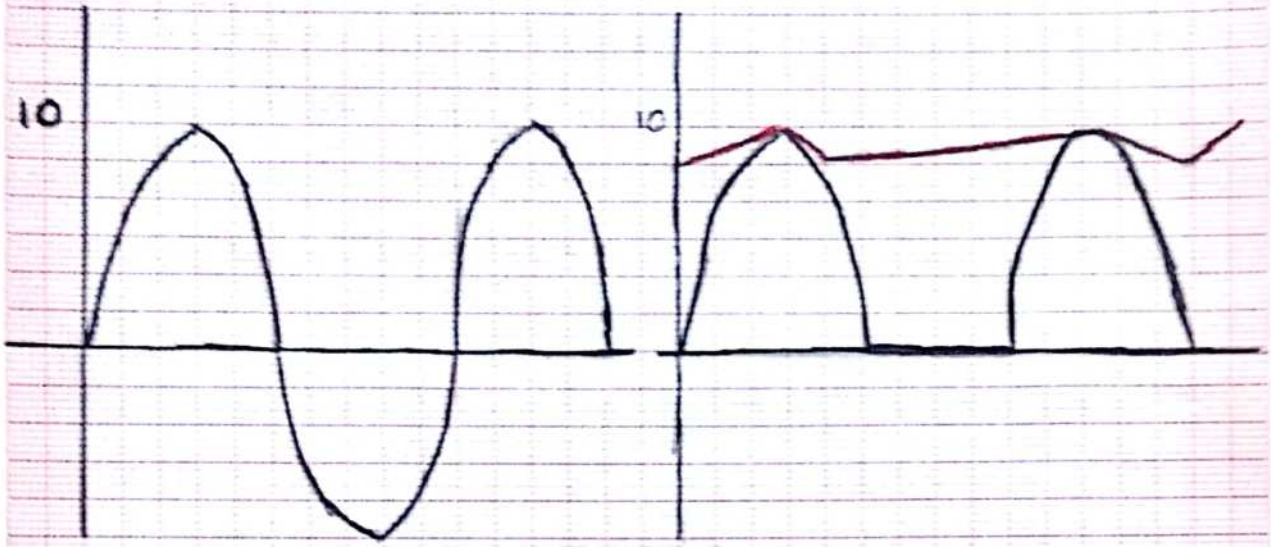
$$r = \frac{V_r}{V_{DC}} = \frac{9.79}{0.59} = 16.6$$

Conclusion :-

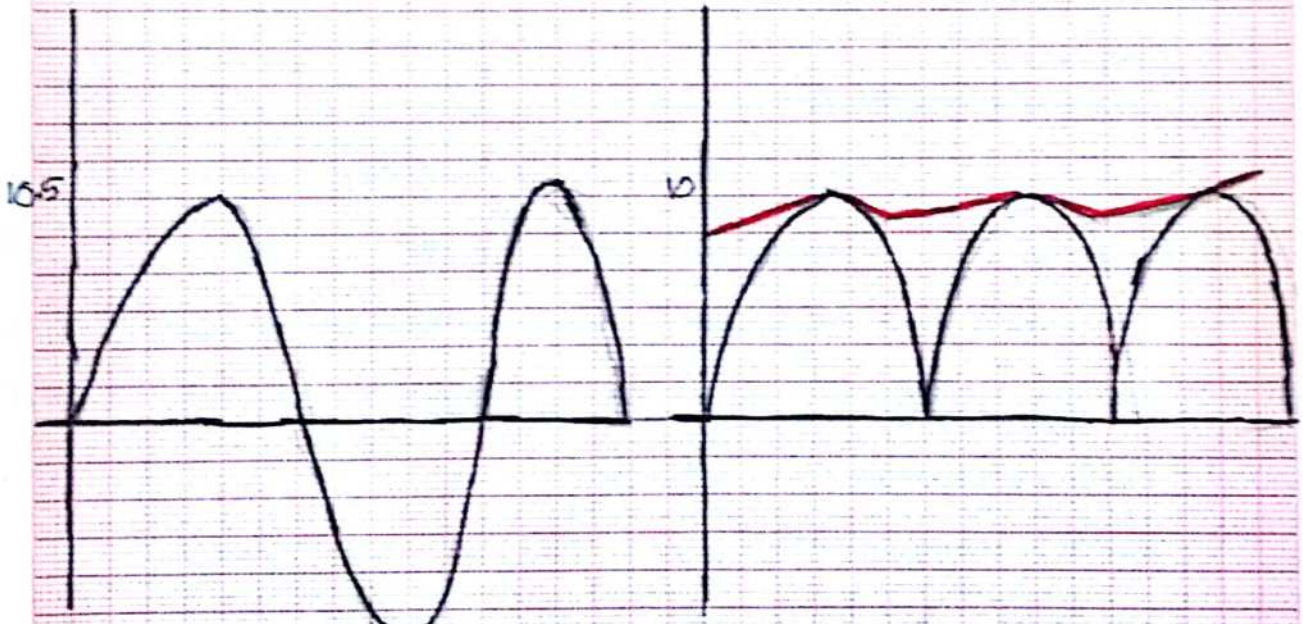
In this experment we learnt about the which experiment have 3 types : Half wave , center topped and Bridge. Also, we studied the benifit of filter which make the Dc wave more straight (less ripples).

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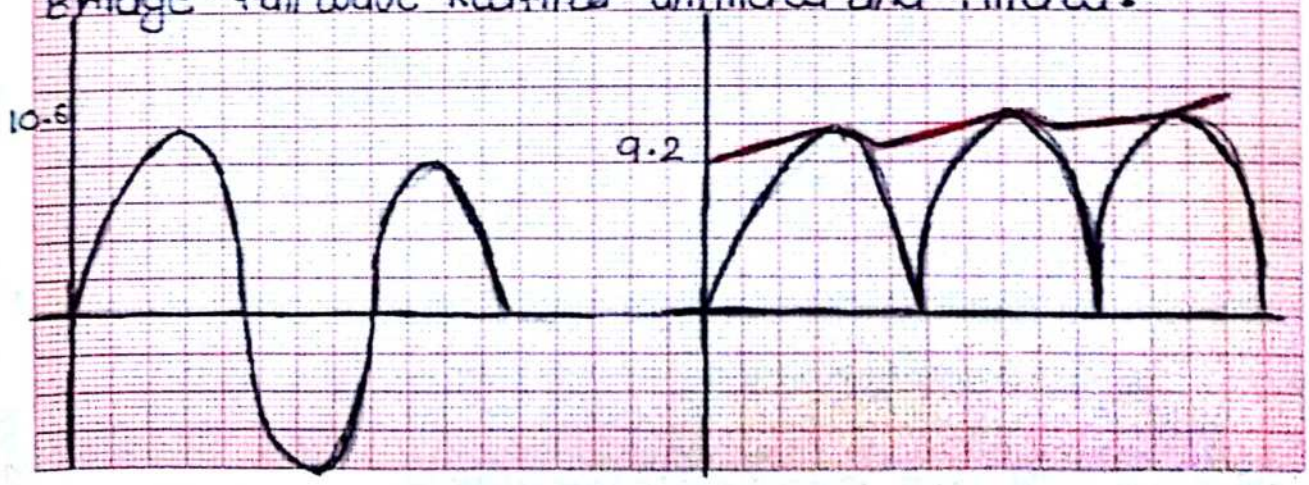
Half wave Rectifier unfilter and filtered :-



Center topped Full wave unfilterd and filterd Rectifier:



Bridge Full wave Redfired unfiltered and filtered :-



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Expereiment Three :-

The diode Limter the diode clamper :-

PART (1)

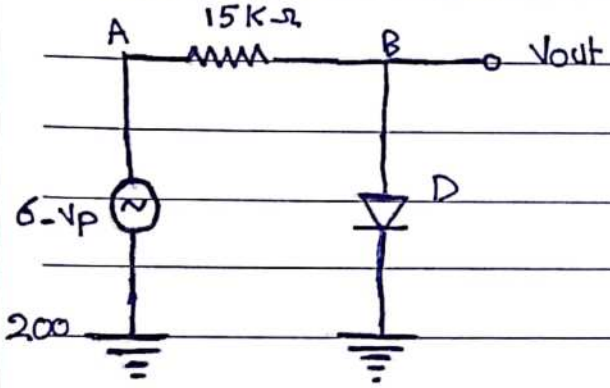


Figure (A)

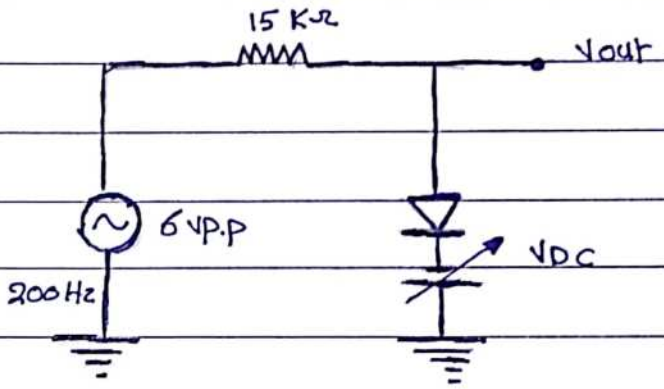
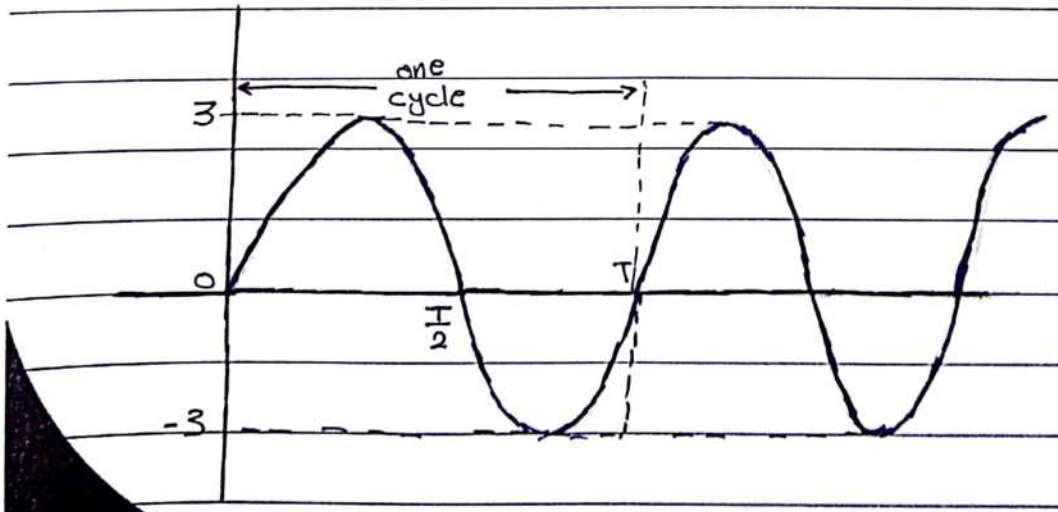


Figure (B)

Vout = VD = 0.7 → positive half cycle.

$V_p = \frac{V_{P-P}}{2} = 3$ volt Vout = VD = -3 volt

$I_D = \frac{V_{in} - 0.7}{R} = \frac{3 - 0.7}{15 * 10^3} = 0.153$ mA



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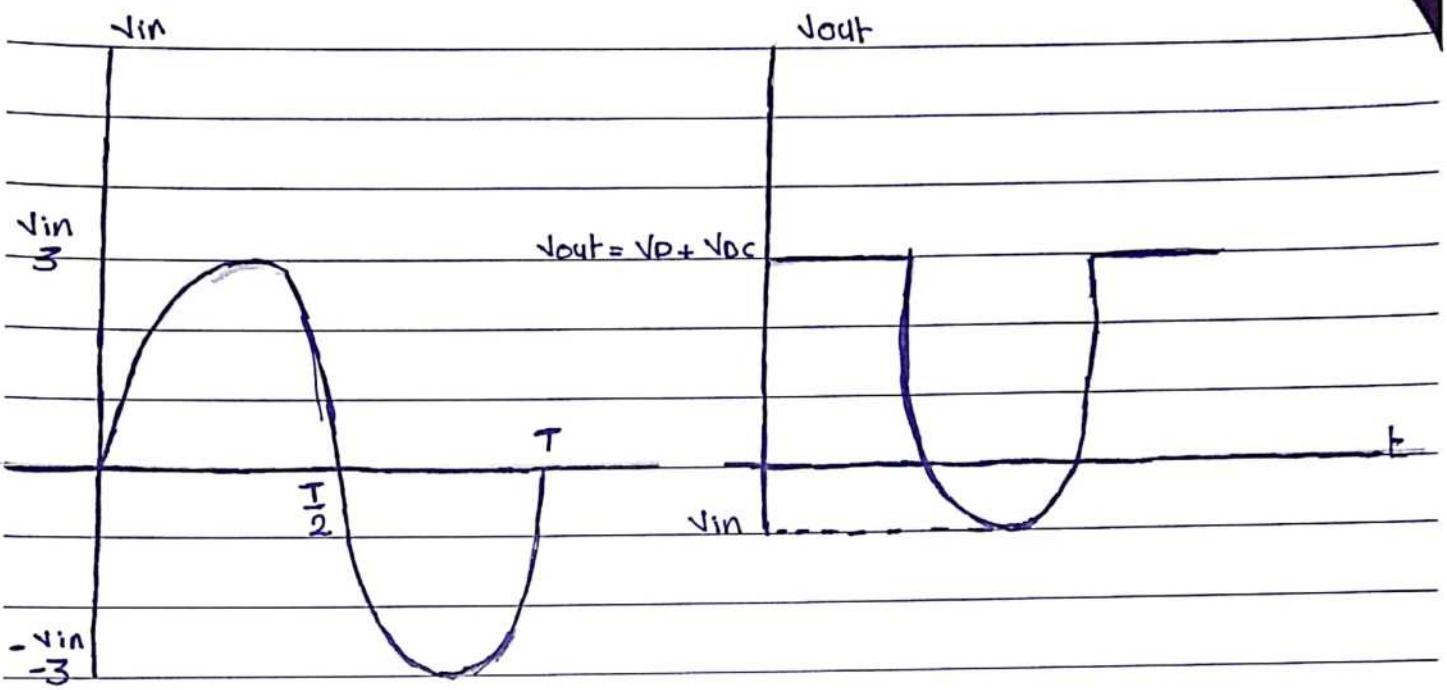
$V_{out} = V_D + V_{DC}$ → positive half cycle

$= 0.7 + \frac{R_2}{R_1+R_2} + V_{CC}$

$V_{out} = V_{in}$ → neg. half cycle.

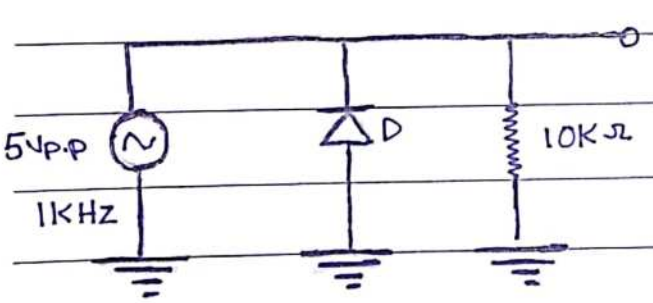
$V_{out} = V_{in} = \frac{V_{P-P}}{2} = \frac{6}{2} = -3$ volt

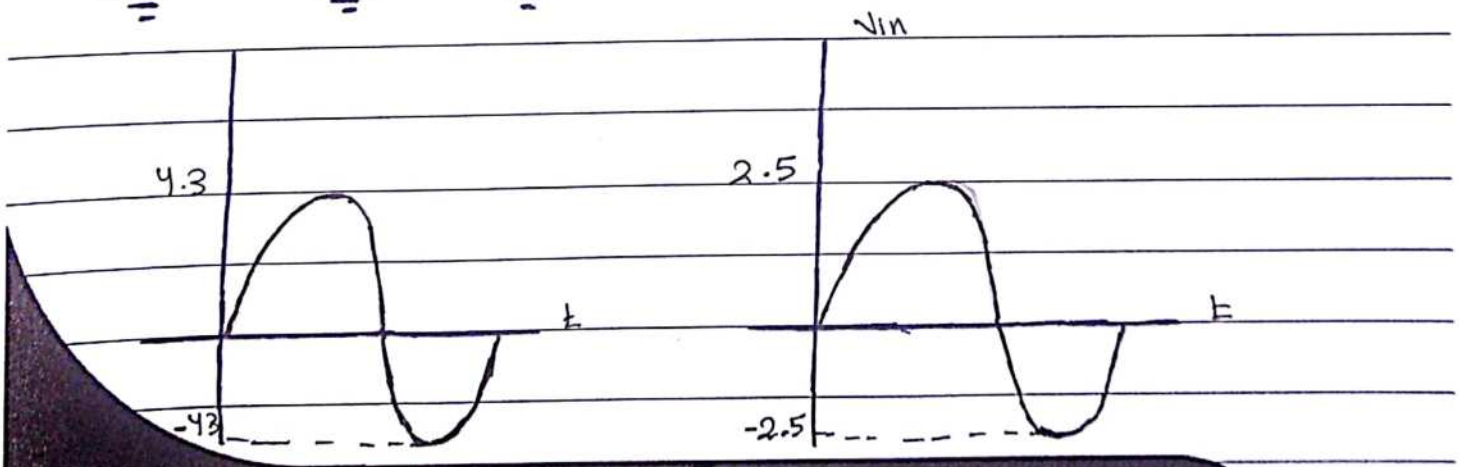
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PART(2)

Clamper circuit





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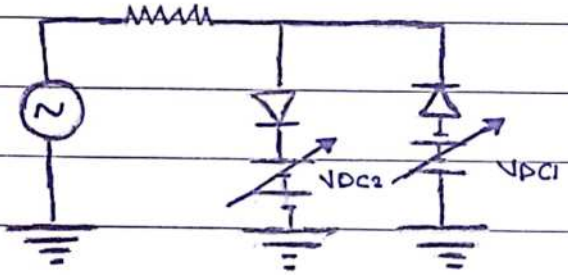
positive half cycle

$V_{out} = V_R = +V_{in} + V_C$

$V_C = \frac{5}{2} - 0.7 = 1.8 \rightarrow V_R = +\frac{5}{2} + 1.8 = 4.3$

negative half cycle

$V_{out} = V_D = 0.7$ volt



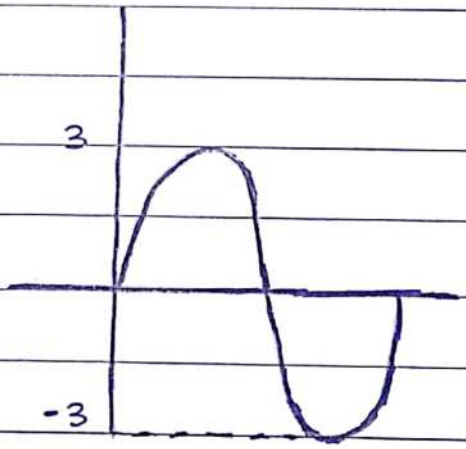
positive half cycle

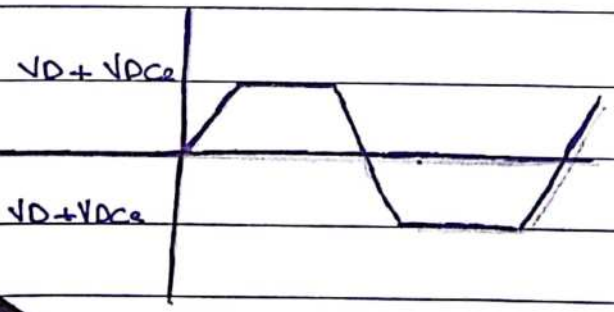
$V_{out} = V_D + V_{DC1}$

negative half cycle

$V_{out} = V_D + V_{DC2}$

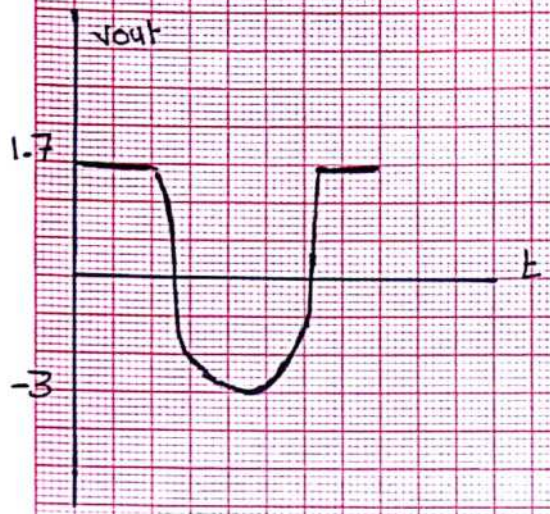
$V_R = V_{in} - (V_D + V_{DC})$



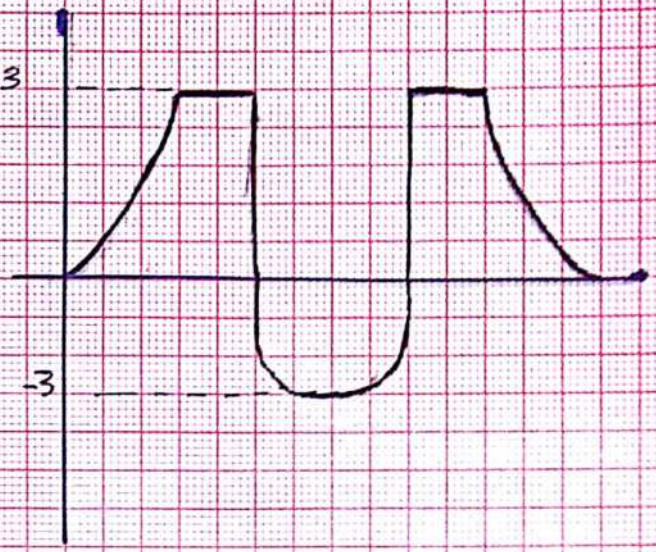


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V_{DC} = 1V



V_{DC} = 2.3



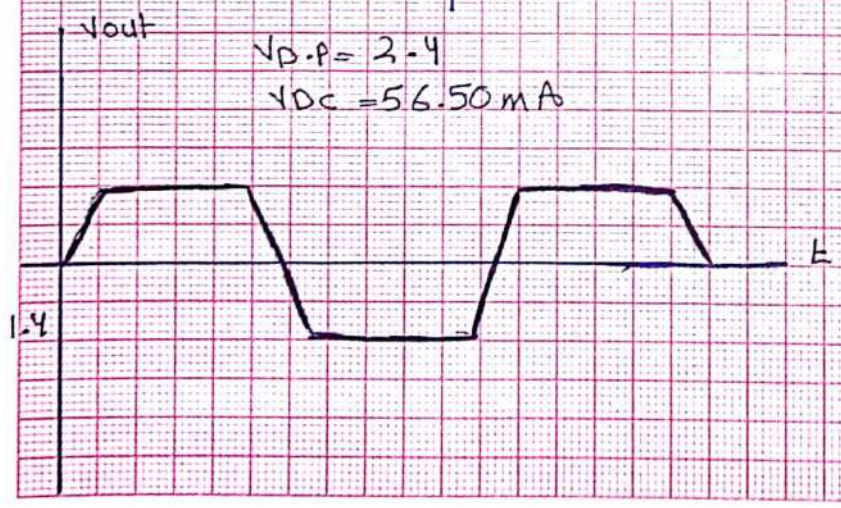
V_{DC} = 8.3



clamper circuit :-



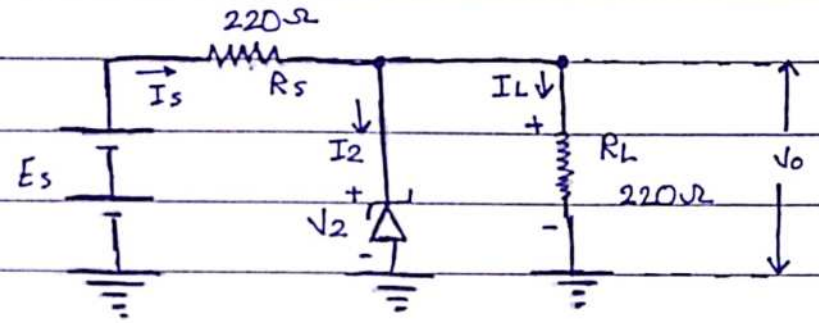
Double Based cliper :-



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Experiement Four:-

Zener characteristics and zener as a Voltage Regulator :-



Figur (1)

Fig(1) is an analysis circuit for zener diodes when are used as avoltage regulators. when Es is less than V_2, zener acts as an open circuit and V_o is given by:

$$V_o = (R_L / R_L + R_s)\, E_s$$

and $I_L = V_L / R_L$

when $E_s \geq V_z$, then

$$V_o = V_L = V_2$$

and $I_s = I_z + I_L$

source current (Is) is given by:

$$I_s = (E_s - V_o) / R_s$$

and power dissipation (P_z) is:

$$P_z = I_z V_z$$

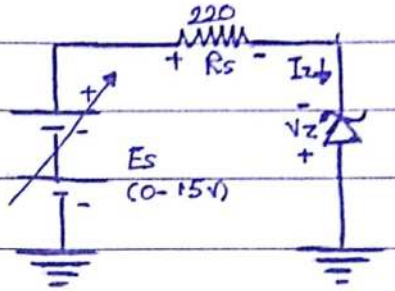
and Regulation percentage (% V.)

$$\% V. = ((V_L - V_{FL}) / V_{FL}) * 100\%$$

when, V_{NL} - output voltage without load, and V_{FL} - output V

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PROCEDURE :-



1. The network is wired.
2. dc supply voltage is increasing in small steps while measuring (Vz) and calculate the current (Iz) using ($I_z = V_{Rs}/R_s$) through the zenerdiode. In the vicinity at zener's knee voltage (approximately) 6V, these steps are mode approximatly 0.05V. a zener current of 40mA is'nt exceeded and data are recorder in table.

$E_s = 5V \rightarrow 5V < 6.2V \rightarrow$ zener is O.C $\rightarrow$

$-5 + I_s R_s + V_2 = 0$

$I_s = 0$

$E_s = 7V \rightarrow 7V > 6.2V \rightarrow -7 + I_s R_s + 6.2 = 0$

$I_z = 0.8/220 = 3.63mA$

$V_z = 6.2V$

Results are plated for the corresponding zener current and voltage is on the graph

4. At $I_2 = 20mA$, determine R_z

$$R_z = \frac{V_z}{I_z} = \frac{6.2}{20 \times 10^{-3}} = 310\Omega$$

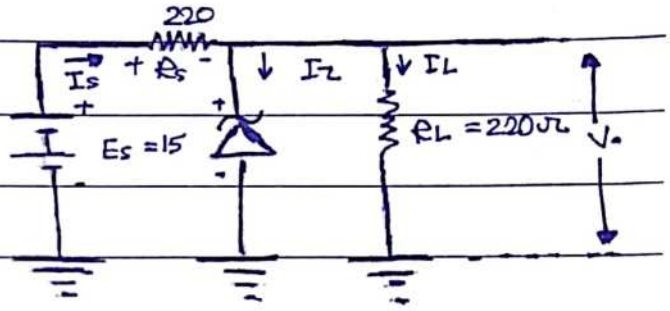
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Es (V)	Vout measure	Vout calc.	Iz measure	Iz calc.
5	4.99	5	0	0
5.5	5.49	5.5	0	0
6	5.91	6	0.36	0
6.05	5.93	6.05	0.55	0
6.15	5.93	6.15	0.75	0
6.2	5.94	6.2	0.95	0
6.25	5.95	6.2	1.16	0
6.3	5.94	6.2	1.4	0.2
6.35	5.95	6.2	1.6	0.45
6.40	5.95	6.2	1.8	0.68
6.45	5.95	6.2	2.0	0.9
6.50	5.95	6.2	2.2	0.13
6.55	5.95	6.2	2.4	1.36
7	5.97	6.2	2.6	1.59
8	5.98	6.2	4.6	3.63
9	5.99	6.2	9.1	8.18
10	5.99	6.2	13.6	12.7
11	6	6.2	18.1	17.2
12	6	6.2	22.7	21.8
13	6	6.2	27.2	26.3
14	6.01	6.2	31.7	30.9
15	6.01	6.2	90.8	80

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PART (2)

Circuit is wired, Is, IL and V. are calculted. Results are recorded in table.



$V_o = V_z = 6.2$

$-15 + 220 I_s + 6.2 = 0$

$I_s = 40\ mA$

$I_L = \frac{6.2}{220} = 28\ mA$

$I_s = I_s - I_L = 40 - 28 = 12\ mA$

2. R_L is Removed, V_o is calculated.

$V_{out} = V_z = 6.2V$

3. Circuit is wired and Is, I_2, I_L and V_o are measured, result are recorded in tabele.

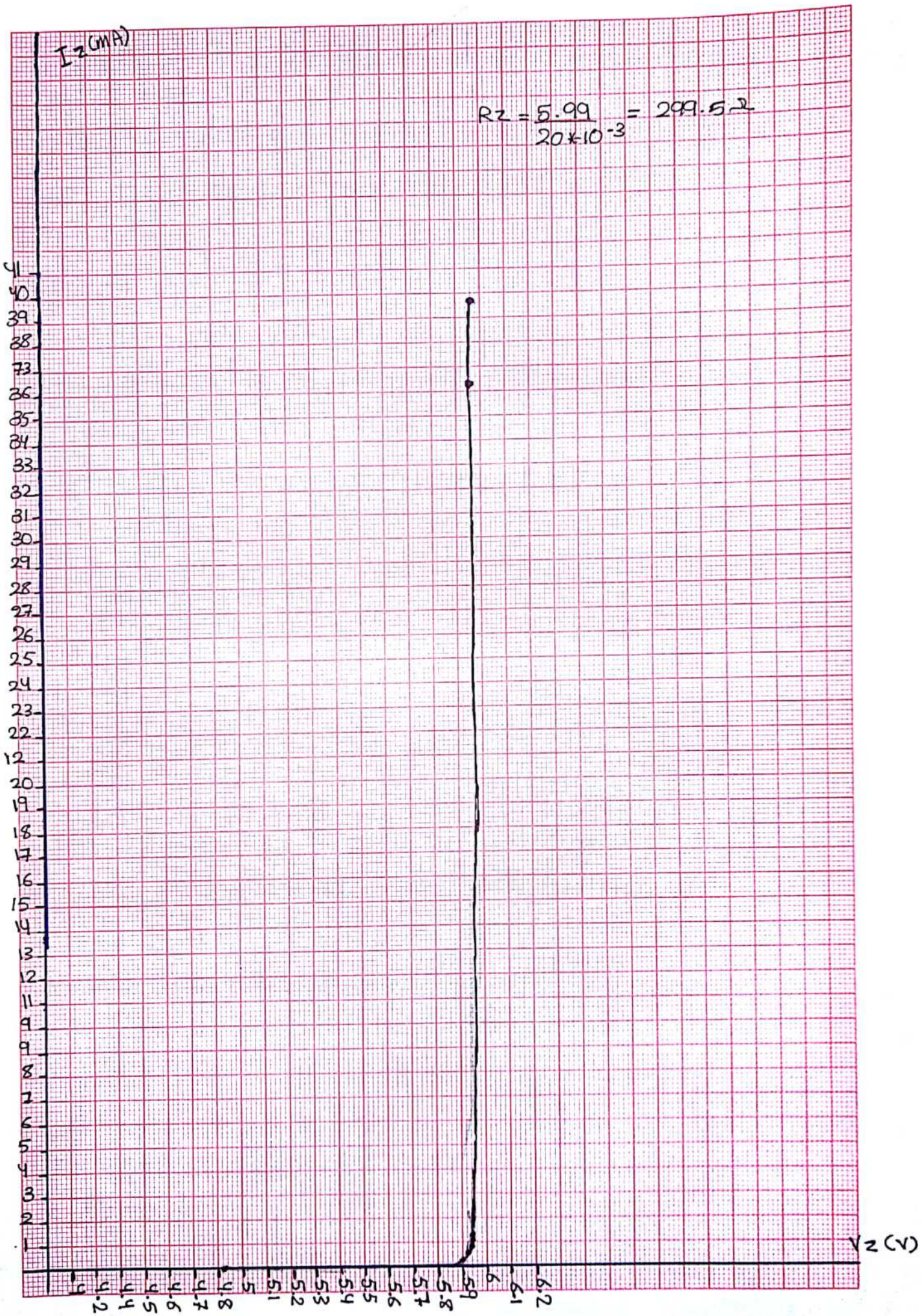
4. RL is removed and V_o is measured

5. Regulation percentage % is calculated.

$\% V = ((V_{NL} - V_{FL}) / V_{FL}) * 100\%$

Parameter	expected value	measurd value	% error
Is	40 mA	39.2 mA	2%
Iz	12mA	11.97 mA	0.25%
IL	2.8mA	27.23	2.75%
V. (with load)	6.2 V	6.12 V	1.29 %
V. (without load)	6.2 V	6.12 V	1.29 %

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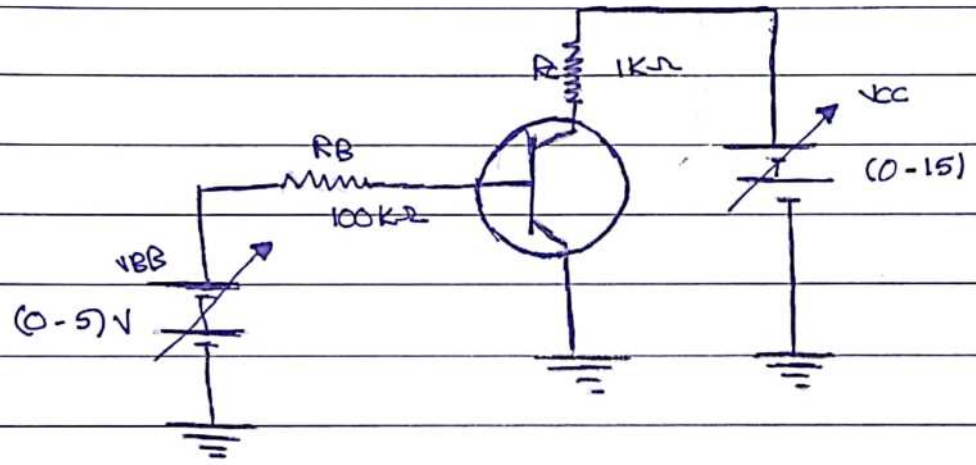
Experiment Five :-

BJT characteristic and self-biased configuration

PART (1)

The collector characteristice.

1. constant the circuit of the figure . Adjust V_{RR} to 1V will seb $I_B = 10mA$ as in Table.



2. set VCE to 2V by varying the Vcc as shown in table

3. Record the VRc and VBE value in table.

$-V_{BB} + R_B I_B + V_{BE} = 0 \rightarrow I_B = \frac{5.07}{100 * 10^3} = 43mA$

$I_C = \beta I_B = 100 * 43 * 10^{-6} = 4.3\ mA$

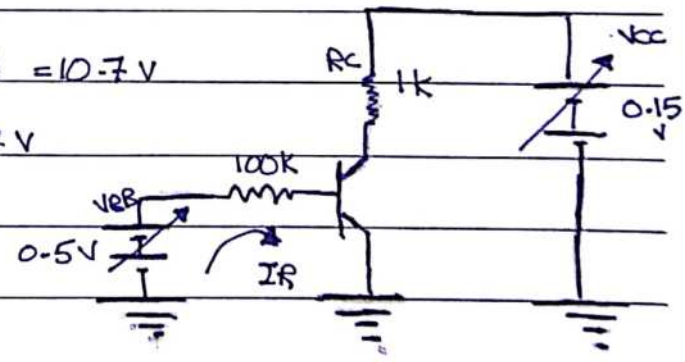
$-V_{CC} + R_C I_C + V_{CE} = 0 \rightarrow V_{CE} = 15 - 4.3 = 10.7\ V$

$V_E = 0 \rightarrow V_{BE} = 0.7 = V_B - V_E \rightarrow V_B = 0.7\ V$

$V_{CE} = 10.7 = V_C - V_E \rightarrow V_C = 10.7$

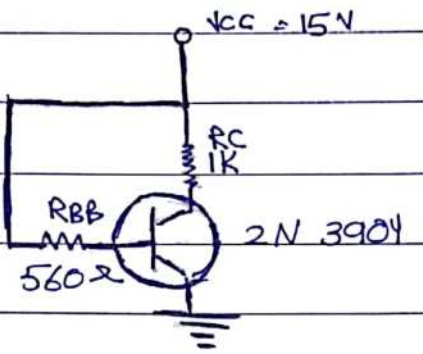
$V_{CB} = V_C - V_B = 10.7 - 0.7 = 10V$

$V_{RC} = 18V$



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(B) Fixed-Biased configuration :-



1. construct circuit as of the figure using 2N3904 transistor and set Vcc = 15 V
2. measure the voltage VB and VRc, and the record them in table
3. calculate the resulting base current, IB and collector current Ic, using the values obtained, find B.

$$-V_{CC} + R_C I_C + V_{CB} = 0 \rightarrow V_C = 15 = R_C I_C$$

$$-V_{CC} + R_B I_B + V_{BE} = 0 \rightarrow I_B = \frac{15 - 0.7}{560K\Omega} = 25.53$$

$$I_C = \beta I_B$$

$$I_C = 5.11 mA$$

$$V_{RC} = I_C * R_C = 5.11 mA * 1K\Omega = 5.11$$

$$V_{CE} = V_{CC} - V_{RC} = 15 - 5.11 = 9.89 (V)$$

measured value :-

IB = 25.97 mA BDC = 200

VRC = 6.5

VCE = 8.61 V

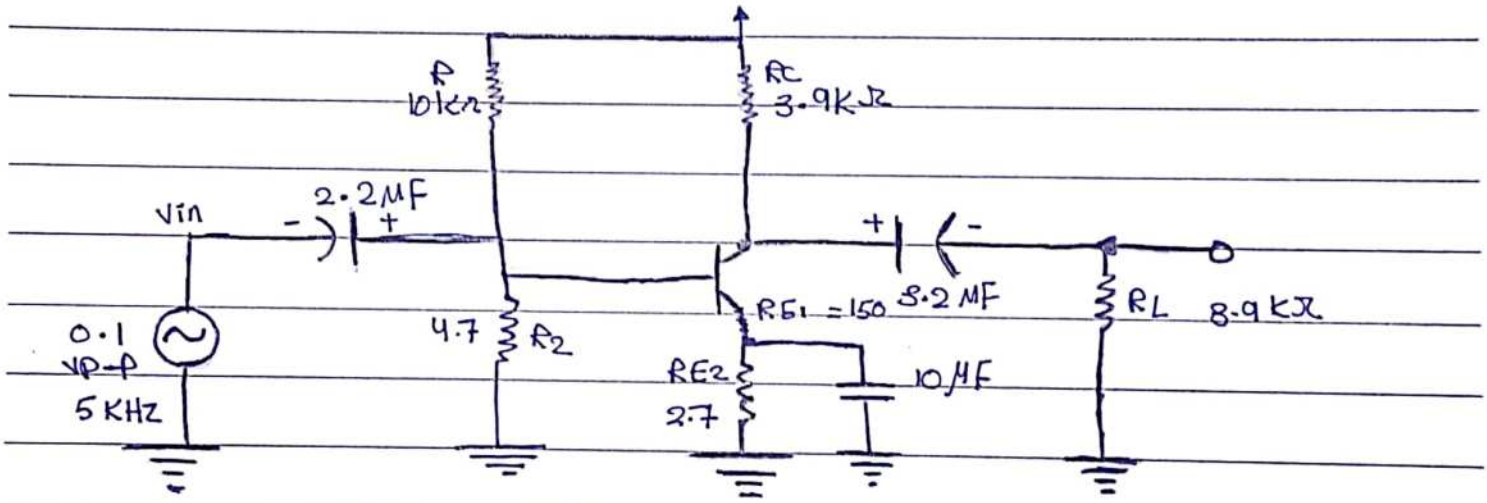
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Expirement six :-

Voltage Divider - Bias configuration and common emitter pimplifier:

PROCEDURE :-

1. assume β for 2N3904 transistor is 200 ; calculate the theoretical values of V_B, V_C, V_E, V_{CE}, I_B, I_C for the network shown, and then recored them in table.



$$V_B = \left(\frac{4.7 * 10^3}{(4.7 + 10) * 10^3}\right)(15) = 4.8V$$

$$V_C = 15 - 3.9 * 10^3 (1.44 * 10^{-3})$$

$$= 9.39V$$

$$I_B = \frac{I_C}{\beta} = \frac{1.44m}{200} = 4.2\,\mu A$$

$$I_C \cong 1.44\,mA$$

$$V_E = V_B - V_{BE} = 4.8 - 0.7 = 4.1V$$

$$V_{CE} = 15 - 1.44 * 10^{-3} (3.9 * 10^3 + 150 + 2.7 * 10^3) = 6.28V$$

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3.

Parameter	expected value	measured value	% error
V_B	4.8V	4.74 V	2%
V_E	4.1V	4.07	0.3%
V_C	9.39V	9.49 V	1%
V_{CE}	5.28V	5.27V	0.18%
I_E	1.44mA	1.43 mA	0.69%
re'	18.05	18.18Ω	0.72%

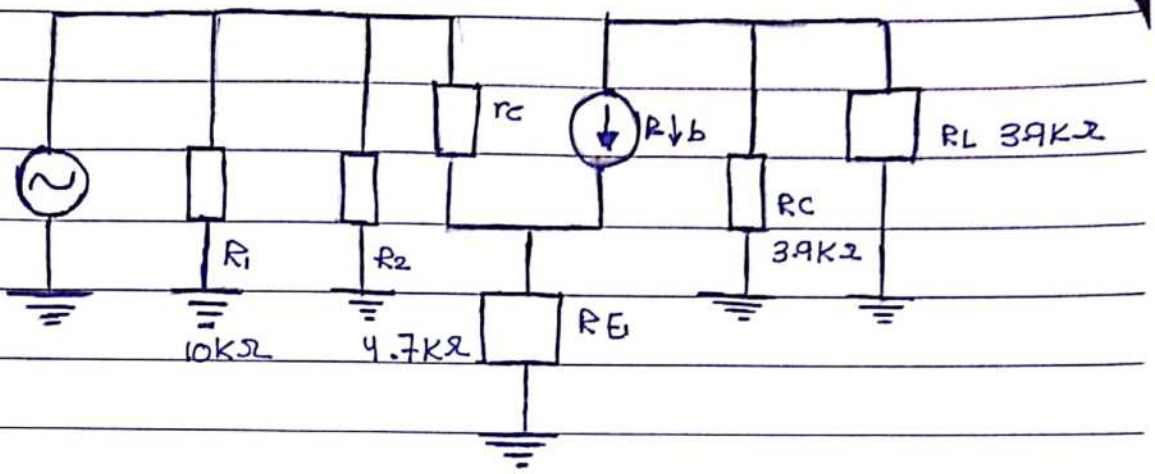
4. set V_{cc} 15V and V_{in} to 0.2 V_{p-p} 1.5KHz

5. Connect CH1 at the input and CH2 at the output. what do you observe? graph the both signal with proper scale on a graph paper.

6. Repeat step 5 fo NO load and NO Bypass capacitor.

condition	V_{in} measure	V_{out} measure	Av expected	measured	% error
normal condition	192.0	2.08	11.6	10.8	6.9%
NO load	191.0	4.16	23.2	21.6	6.9%
NO Bypass cap.	192.0	206	0.68	1.04	5.29%
Fin = 100 Hz	191.0	1.32V	—	6.875	—
Fin = 100KHz	192.0	2.12	—	11.04	—
R_L = 100	—	—	0.65	—	—

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$I_B = 7.2\ MA$

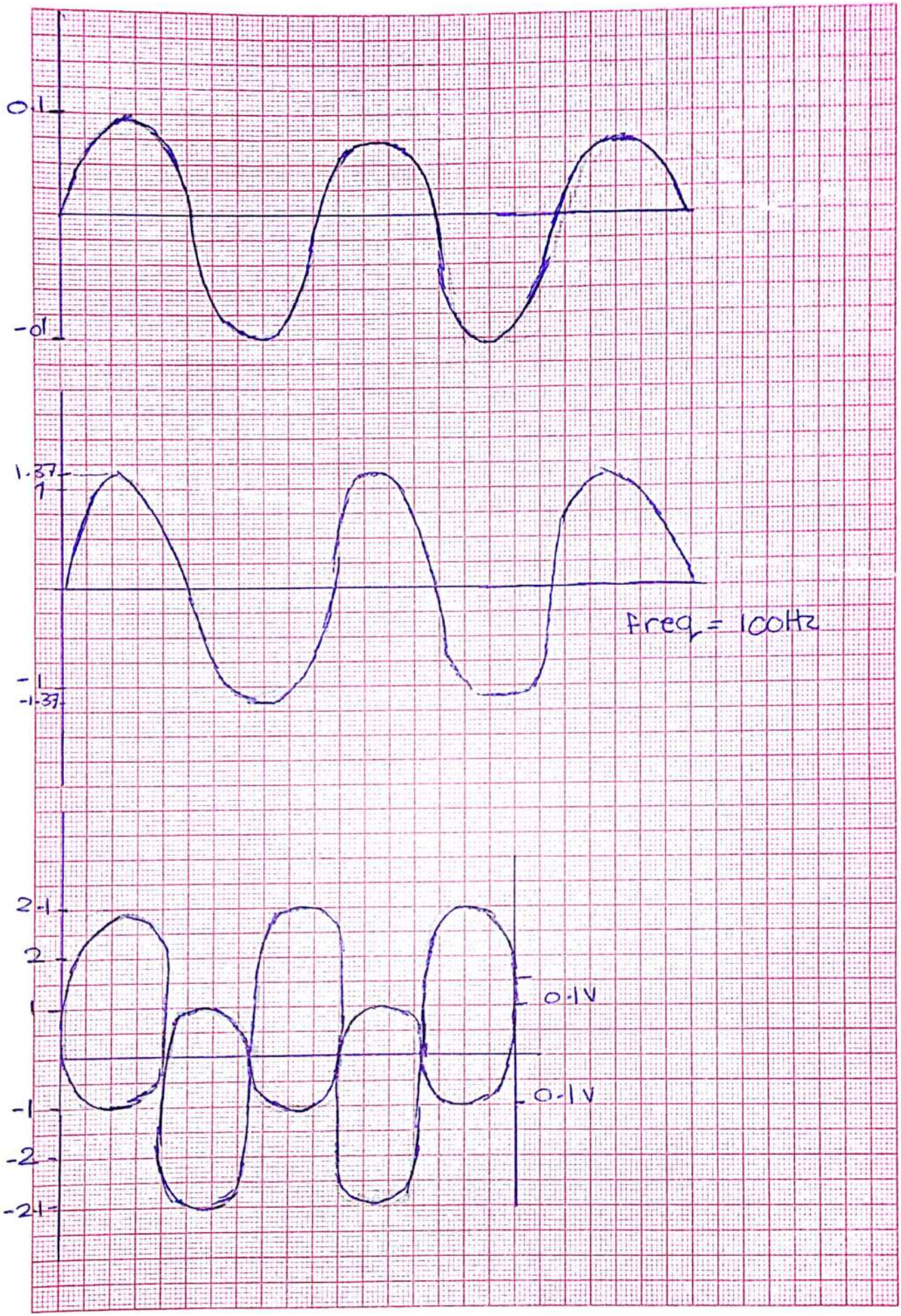
$I_E = 1.44mA$

$V_o = 2.808V$

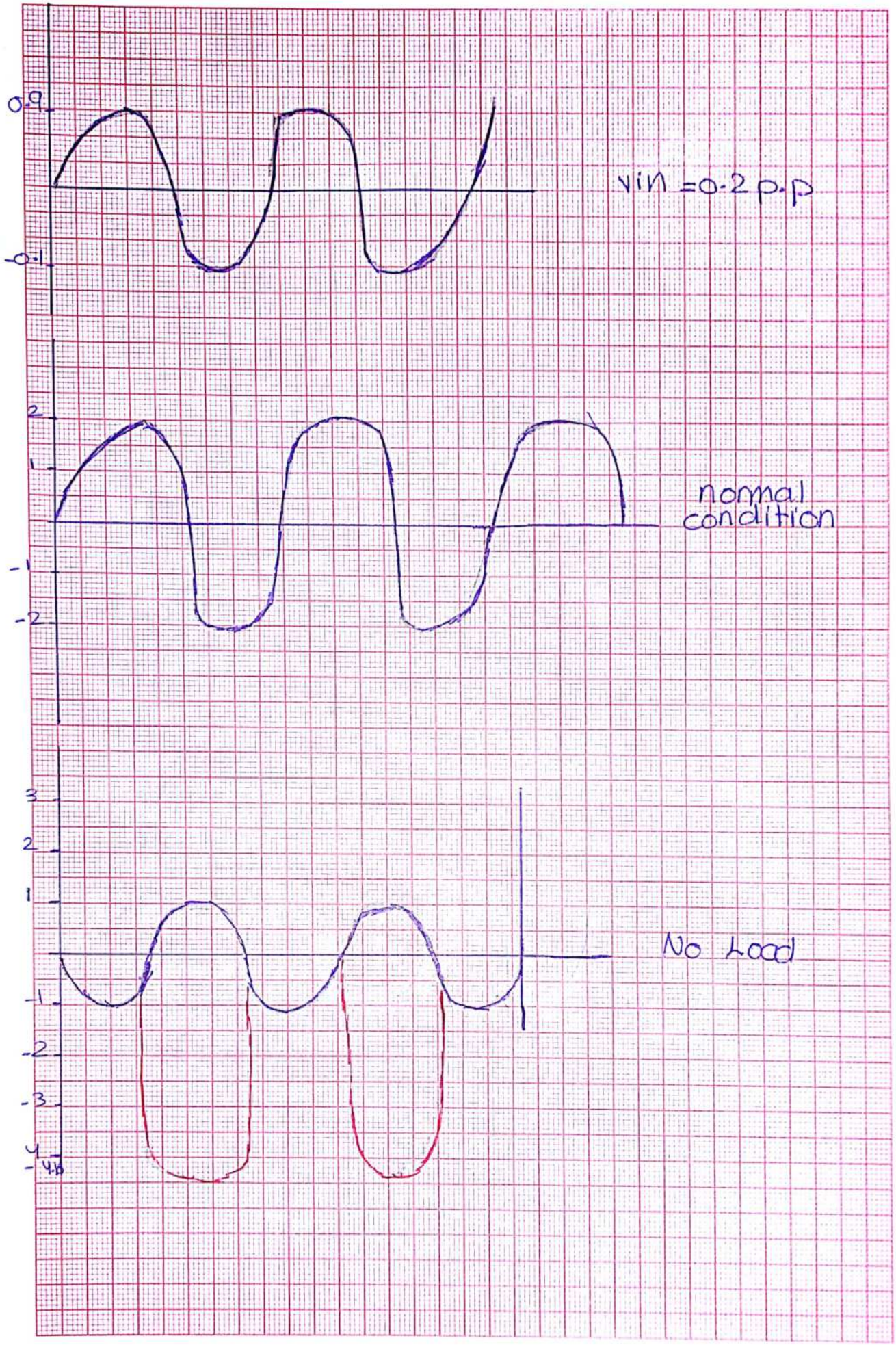
$V = 0.24V$

$A_N = \frac{V_o}{V_i} = \frac{2.808}{0.24} = 11.7$

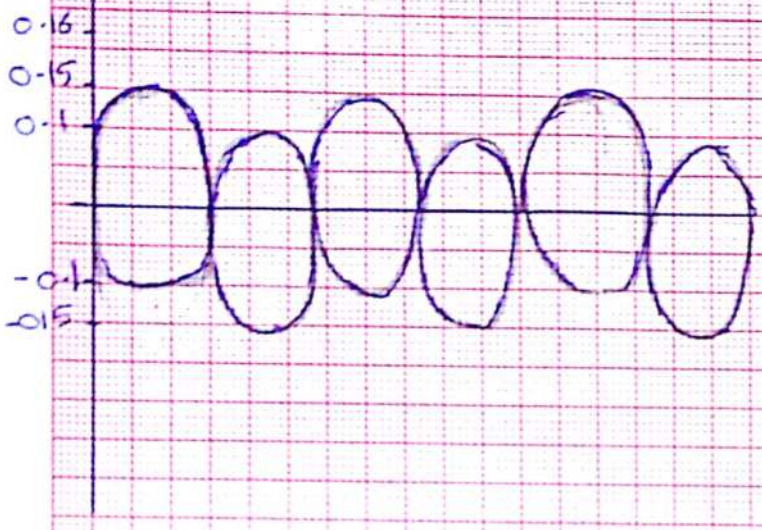
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No Bypass capacitor

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Experiment eight :-

OP-AMPS coperational Amplifiers

PART (1)

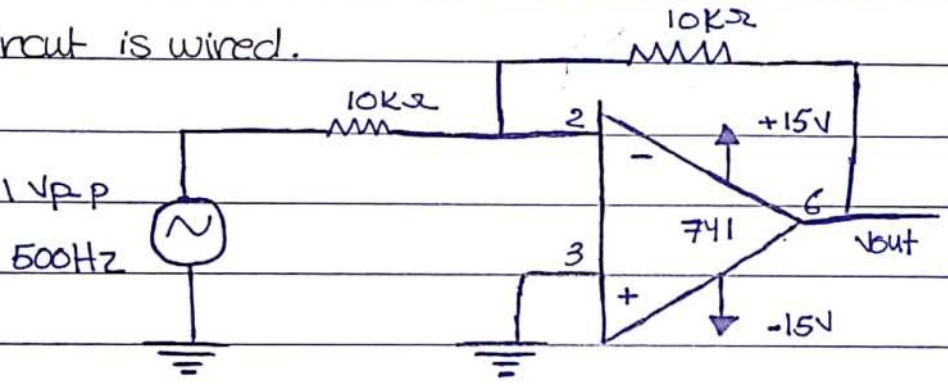
inverting and non inverting Amplifiers

1. expected voltage gain Av is calculated.

$$Av = 10K\Omega = \frac{-Rf}{Ri} = \frac{10K}{10K} = -1$$

$$Av\ (1K\Omega) = \frac{-Rf}{Ri} = \frac{1K}{10K} = -0.1$$

2. Circuit is wired.



3. F.G is adjvsted to 0.5 p-p 1500 Vp-p . The peak-to peak outpat voltage are measured . Voltage gain is determine and compared with the expected .

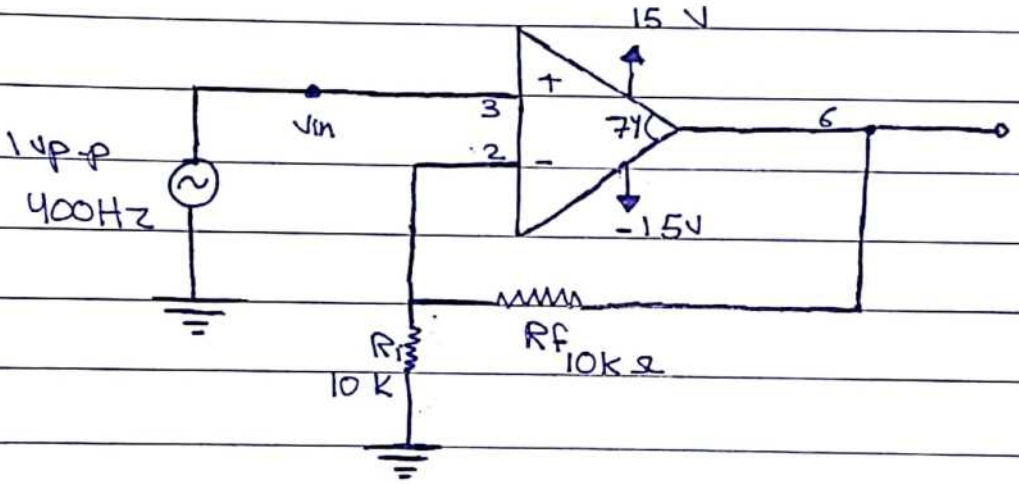
$$Av = \frac{Rf}{Ri} = -1 \quad \text{(expected)}$$

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4. keeping the input signal at 1V peak to peak, Resister or Rf is changed according to table. "Each time, disconnect the power supply and signal generator before you change the resistor"

Rf	measured vout	measured gain	expected gain	-/- error
10KΩ	460	0.96	-1	4-/-
4.7KΩ	—	—	-0.47	—
100 KΩ	—	—	-10	—
1KΩ	9.21	9.21	-0.1	8-/-

5.



Rf	measured vout	measured gain	Expected gain	-/-error
10K Ω	1.88	0.98	2	6-/-
4.7K Ω	—	—	1.47	—
100K Ω	—	—	11	—
1KΩ	9.76	9.76	1.1	11.2-/-

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PART (2)

OP-AMP comparators :-

The purpose of this part is to demonstrate the operation of noninverting and inverting comporator circuit using a 741 operational amplifier. A comparator circuit determine whether an inpat voltage is greater than apedetemine refrence level. since a comparator operates in an open-loop mode, the output voltage opproaches either its positive or its negative supply voltage.

useful formulas :

Noninverting comparator output

(1) $V_{out} = +V_{sat}$ when $V_{in} > V_{Ref}$

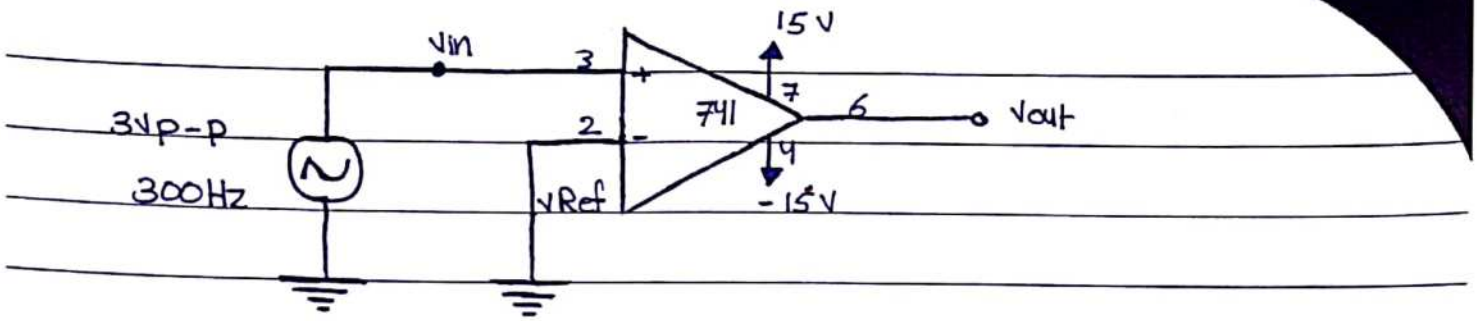
(2) $V_{out} = -V_{sat}$ when $V_{in} < V_{Ref}$

Inverting comparator outpat

(3) $V_{out} = +V_{sat}$ when $V_{in} < V_{Ref}$.

(4) $V_{out} = -V_{sat}$ when $V_{in} > V_{Ref}$.

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2. power applyed HD the breadboard 3Vp-p voltage is adjusted and freqyancy at 300 Hz. what is the polority of the output voltage when the input signal goes positive? when the input goes negative?

$V_{in} > V_{Ref}$, $V_o = 15V$

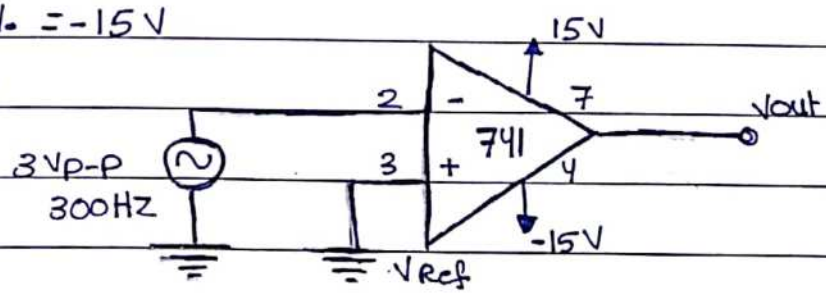
$A_V = \frac{15}{3} = 5$

3. power and signal leads to the breadboard are disconnected and the input connections to the op-amp are reversed so that the input signal is connected to the inversting inpat while noninversting input is grounded.

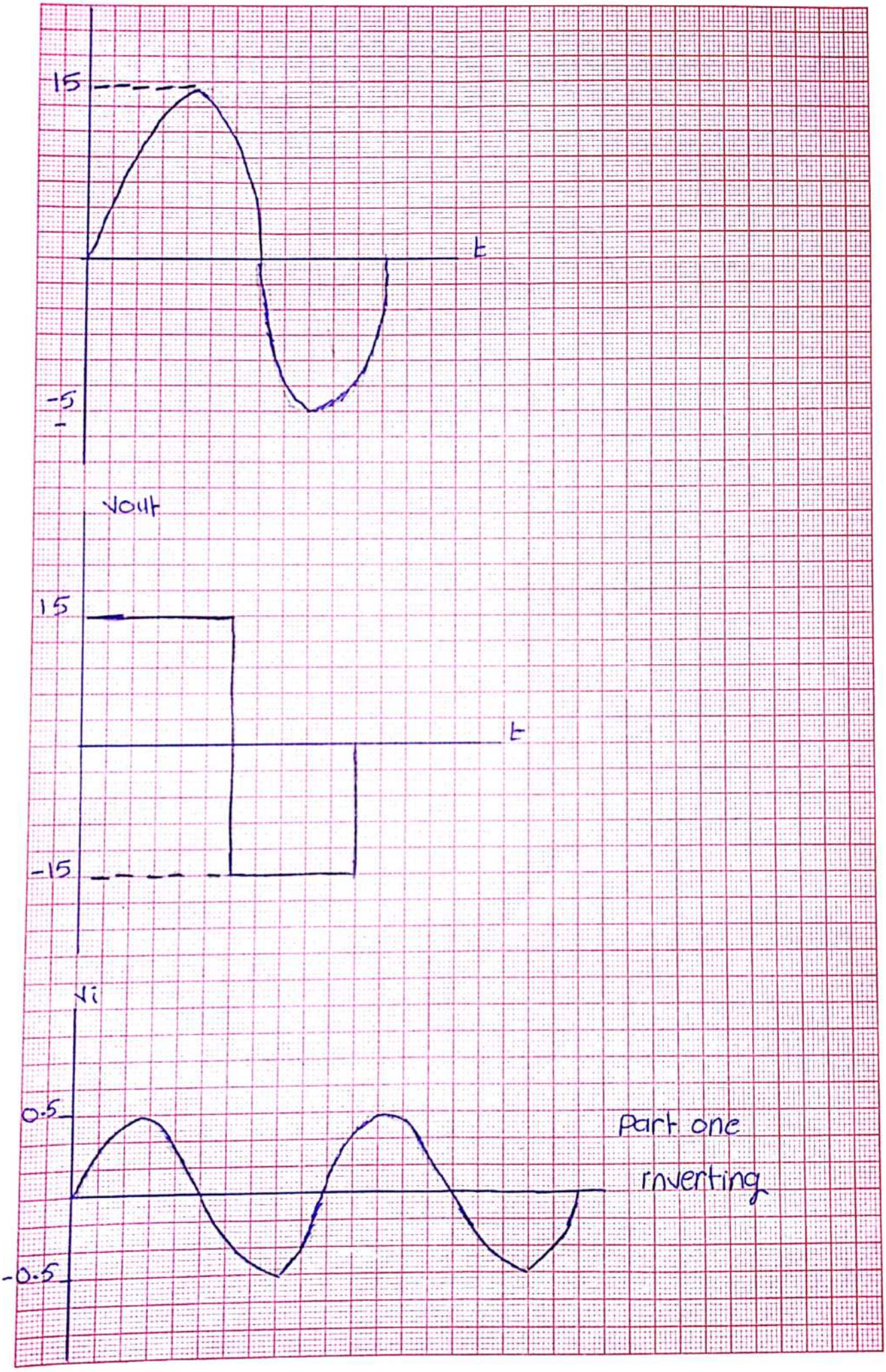
4. power and signal leads to the breadboard are applyed. what is the difference between the operation of this circuit and that of the circuit used earlier?

$V_{in} > V_{Ref}$, $V_o = -15V$

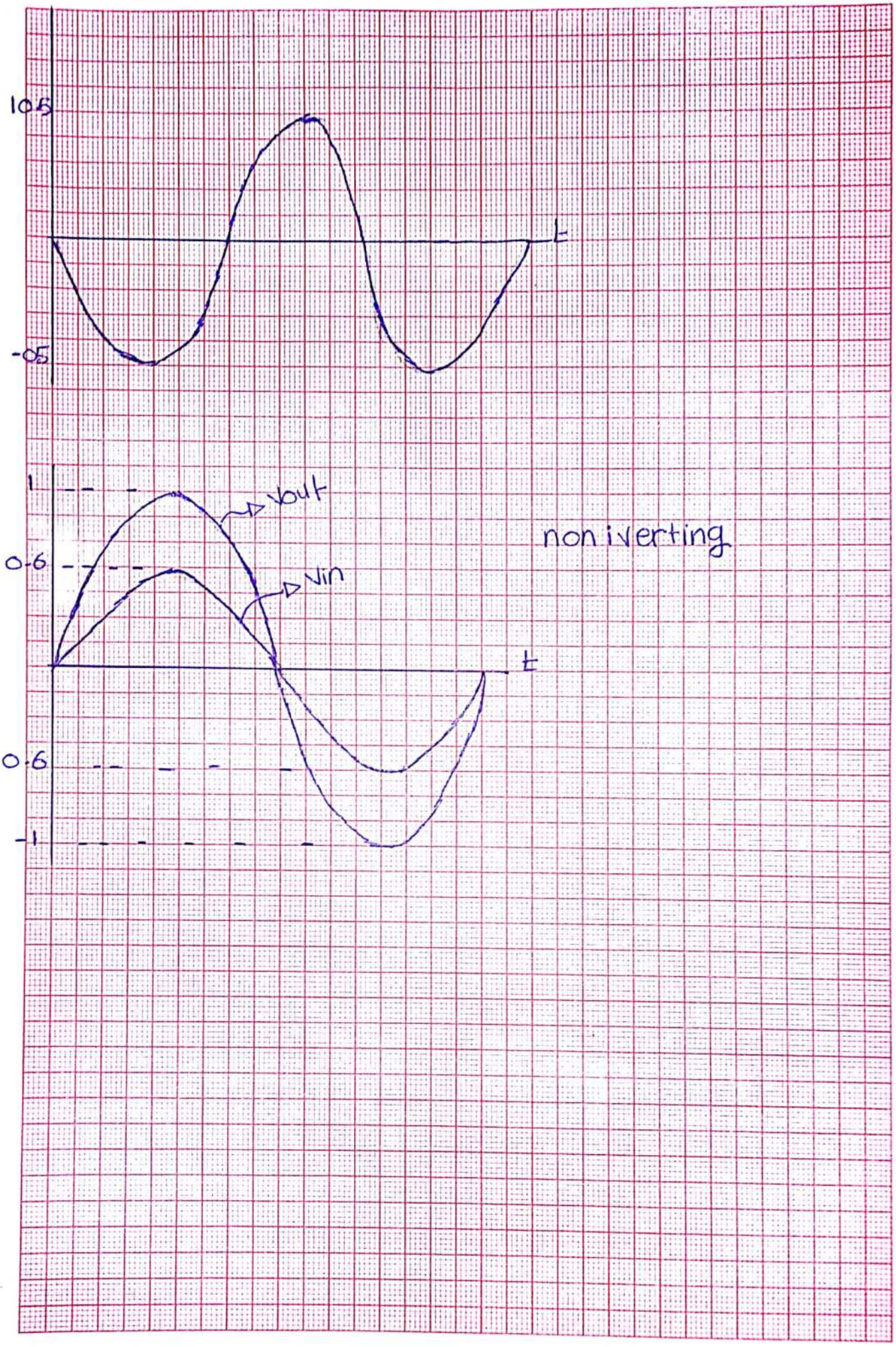
$A_V = \frac{-15}{3} = -5$



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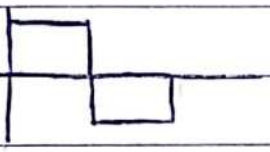
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Experiment nine

OP-AMP differetiator and Integrator :-

1. wire the differentiater circuit shown in figur and set your oscilloscope to the following approximate
   channel 1 : 0.5 V1 division dc coupling
   channel 2 : 0.05 V1 division , dc coupling
   Time base : 0.5 ms/ division

2. Apply power to the bread board , and adjust the peak-to-peax voltage of the input triangle wave at 1V and the frequency at 400Hz . what is shape of the output signal ?



Rectangular

3. Temporaily remove the probe connected to channel 2 the oscillo scope , and adjust the resulting straight Line (ground level) at same convient position on the screen . Reconnect the pobe the output of the differentiator and measure the negative peak voltage (with respect to ground) of the squar wave, recording your result in table 10.1

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4. change the time base to 0.2 ms/division and channel 2 to 0.1 division. Then adjust the input frequency at 1KHz Repeat steps 2 and 3. you should find that peak output voltage increases.

5. Now change the input frequency to 30 KHz. Adjust the time base to ms/division and channel 2 to 2 v/ division. what does the out put signal look like?

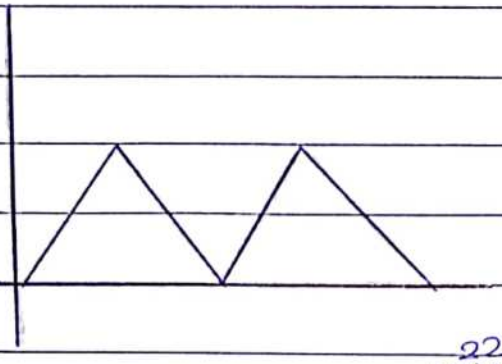
6. Measure the peak-to-peak output voltage and determine the voltage gain, recording in table 10.1. How does the voltage gain compare to that of an inverting amplifier?

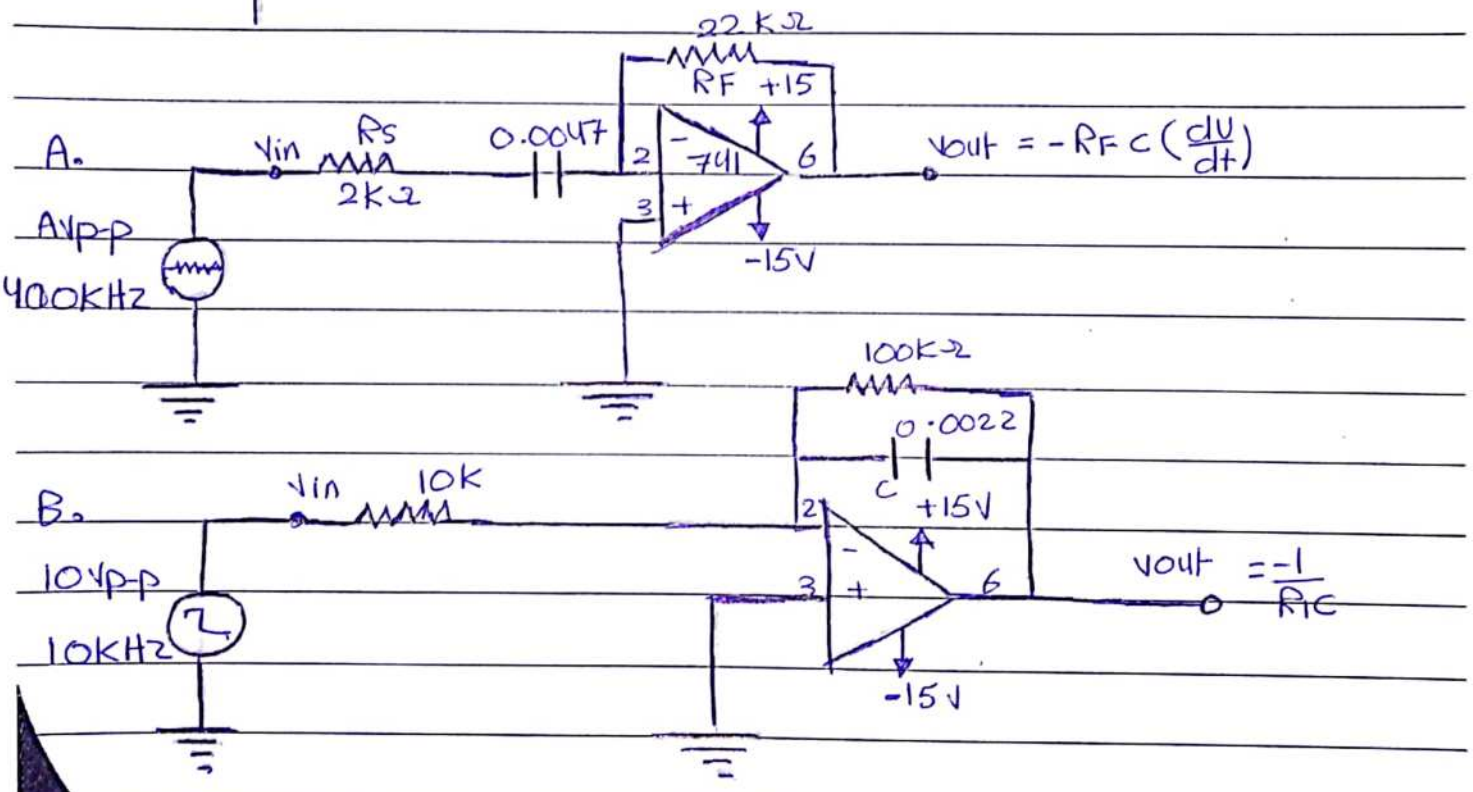
7. wire the integrator circuit shown in the shematic digram of figure 10.1. and set your oscilloscope to the following approximate setting:

chanels 1 and 2 : 0.5 v/ division, dc coupling Time base 20 Ms/ division.

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8. Apply power to the breadboard, and adjust peak-to-peak voltage of the input squar wave at 1V and the frequency at 10 KHz. How does the output signal? Draw the waveforms?
Record your result in table 10.2





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Figure 10.1 : a) Differentiator Circuit
b) Integrator

10.1 : differentiator

input frquency	peak output
400Hz	
1K Hz	
30K Hz	

10-2 : Integrato:

input frequency	peak output
10KHz	
4 K Hz	
100Hz	

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$$f_c = \frac{1}{2\pi R_S C} = \frac{1}{2\pi * 2 * 10^3 * 0.0047 * 10^{-6}} = 16.9\,KHz$$

$$X_C = \frac{1}{2\pi f_c} = \frac{1}{2\pi * 400 * 0.0047} = -.84656.88$$

$$R = \sqrt{(2^2 + 84^2) * 10^6} = 84\,K\Omega$$

$$V_o = -RFC\left(\frac{dv_{in}}{dt}\right) = -22k * 0.0047 * 10^{-6} * 2 = 20.68mV$$

$$A_V = \frac{V_o}{V_{in}} = \frac{20.68\,mV}{1} = 0.2$$

$$f_c = \frac{1}{2\pi R_S C} = \frac{1}{2\pi * 2 * 10^3 * 0.0047 * 10^{-6}} = 16.9\,KHz$$

$F_{in} = 1KHz \rightarrow F_{in} < F_c$

$$X_C = \frac{1}{j2\pi f_c} = -j\,84656.88 \rightarrow R = \sqrt{(2^2 + 84^2) * 10^6} = 84K\Omega$$

$$R = 84K\Omega \rightarrow A_V = \frac{V_o}{V_{in}} = \frac{20.68}{1} = 0.2\,mV$$

$f_c = 16.9Hz \rightarrow F_{in} = 30KHz > F_c$

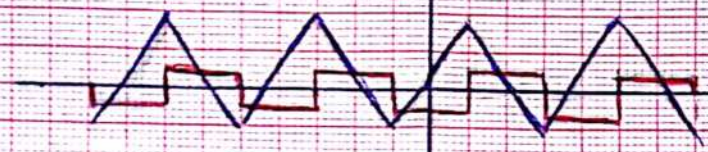
the circuit approaches an inverting amplifier with voltage

$$\frac{-R_F}{R_S} = \frac{-22\,K\Omega}{84\,K\Omega} = 0.26$$

$$V_o = A_V \cdot V_{in} = 0.26 * 1 = -260\,mV$$

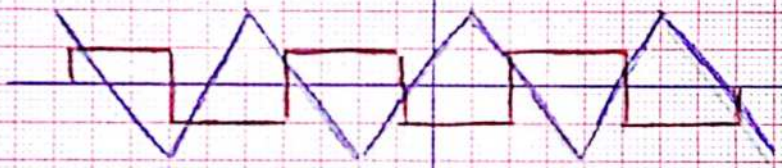
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Fin = 400Hz



1ms
I/P ← 500mV
O/P ← 50mV

Fin = 1000 Hz



1ms
I/P ← 50mV
O/P ← 50mV

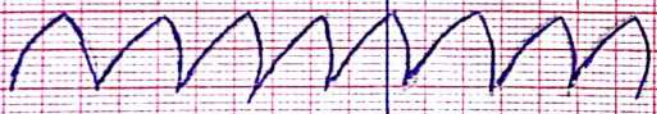
Fin = 30 KHz

500mV

50μs



500mV



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